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%pip install CoolProp

import numpy as np
import matplotlib.pyplot as plt
import CoolProp.CoolProp as CP

def plot_Pv_diagram(fluid='Water'):
    print(f"Generating P-v Diagram for {fluid}...")

    # 1. Get Critical Point Data
    T_crit = CP.PropsSI('Tcrit', fluid)
    P_crit = CP.PropsSI('Pcrit', fluid)

    # 2. Generate the Saturation Dome (Liquid and Vapor lines)
    # We span temperatures from Triple Point up to Critical
    ↪ Point
    T_min = CP.PropsSI('Tmin', fluid)
    T_vals = np.linspace(T_min, T_crit - 0.1, 100) # Stop just
    ↪ before critical point

    v_liq = []
    v_vap = []
    P_sat = []

    for T in T_vals:
        # Calculate Saturation Pressure at this T
        P = CP.PropsSI('P', 'T', T, 'Q', 0, fluid)
        P_sat.append(P)

        # Calculate Saturated Liquid Volume (Q=0)
        rho_l = CP.PropsSI('D', 'T', T, 'Q', 0, fluid)
        v_liq.append(1/rho_l)

        # Calculate Saturated Vapor Volume (Q=1)
        rho_v = CP.PropsSI('D', 'T', T, 'Q', 1, fluid)
        v_vap.append(1/rho_v)

    # 3. Generate Isotherms (Constant Temperature lines)
    # Let's plot 3 isotherms: One below crit, one at crit, one
    ↪ above
    isotherms_C = [300, 373.946, 450] # Temperatures in Celsius
    # (Note: Water T_crit is approx 373.946 C)

    plt.figure(figsize=(8, 6))

    # Plot Dome
    plt.plot(v_liq, P_sat, 'b-', label='Saturated Liquid', lw=2)
    plt.plot(v_vap, P_sat, 'r-', label='Saturated Vapor', lw=2)

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# Plot Isotherms
v_iso = np.logspace(-3, 1, 100) # Log spacing for volume is
↪ better

for T_c in isotherms_C:
    T_k = T_c + 273.15
    P_iso = []
    for v in v_iso:
        try:
            # Calculate P given T and v (Density)
            p_val = CP.PropsSI('P', 'T', T_k, 'D', 1/v,
↪ fluid)
            P_iso.append(p_val)
        except:
            P_iso.append(None)

    plt.plot(v_iso, P_iso, '--', label=f'T = {T_c} C')

# Formatting to match textbook style
plt.xscale('log') # Crucial for P-v diagrams
plt.yscale('log')
plt.xlabel('Specific Volume $v$ ($m^3/kg$)')
plt.ylabel('Pressure $P$ (Pa)')
plt.title(f'Log-Log P-v Diagram for {fluid}')
plt.grid(True, which="both", ls="--", alpha=0.3)
plt.legend()
plt.axhline(P_crit, color='k', linestyle=':',
↪ label='Critical Pressure')

plt.show()

# Run the function
plot_Pv_diagram('Water')

```

Collecting CoolProp

Downloading coolprop-7.2.0-cp313-cp313-macosx_11_0_arm64.whl.metadata (4
Requirement already satisfied: numpy>=1.20 in /opt/anaconda3/lib/python3.13
packages (from CoolProp) (2.3.4)

Downloading coolprop-7.2.0-cp313-cp313-macosx_11_0_arm64.whl (5.6 MB)

5.6/5.6 MB 42.0 MB/s eta 0:00:00

Installing collected packages: CoolProp

Successfully installed CoolProp-7.2.0

Note: you may need to restart the kernel to use updated packages.

Generating P-v Diagram for Water...

